

TAPAS KAR

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Summary

Back-end & Cloud Software Engineer with experience at Amazon Alexa and Rimidi, specializing in Java/Spring Boot, Python/Django, REST APIs, SQL/Oracle databases, and distributed systems. Skilled in AWS, Docker, Kubernetes, Kafka, CI/CD pipelines, and Linux/Unix/Shell scripting, with additional frontend exposure (Vue.js, React). Proven track record of delivering cloud-native, scalable, and resilient production systems with monitoring and test automation. *H-1B already approved - new employer only needs to file a transfer.*

Education

Arizona State University

Master of Science in Robotics and Autonomous Systems - Artificial Intelligence

Aug. 2020 – May 2022

Tempe, AZ

The University of Texas at Tyler

Bachelor of Science in Computer Science and Mathematics

Aug. 2016 – May 2020

Tyler, TX

Work Experience

Rimidi, Inc.

Software Engineer - Product Support

May 2023 – Present

Atlanta, GA

- Implemented troubleshooting, maintenance, and end-to-end product features across Python/Django backends and Vue/JavaScript front-end, deploying via Docker containerization on AWS ECS integrated with CI/CD pipelines (GitHub Actions, Jenkins)
- Designed and documented cloud-native monitoring solutions using AWS CloudWatch alarms (Terraform IaC), Grafana, and Slack integrations, improving visibility into CPU/memory health and cutting incident response times by 30%
- Partnered with cross-functional QA to validate product lifecycle and quality, analyzing test outputs, automating workflows, and debugging REST/FHIR API integrations with Postman and SQL queries, ensuring compliance with user requirements
- Developed full stack Clinical Decision System UI called "Cards", fetching and rendering recommendations with accept/deny actions and Epic EHR integration, streamlining provider workflows
- Resolved 50+ Jira issues per quarter by analyzing uWSGI, HAProxy, and Nginx logs, shortening resolution cycles by 35% and stabilizing operationally critical systems
- Supported new business growth by collaborating on RESTful FHIR APIs/EHR integrations (Epic, eCW, Athena) and device APIs (Dexcom, Libre, Continua) using AWS Lambda, enabling Rimidi to onboard 3 healthcare clients

Tech: Python, Django, HTML5, CSS, JavaScript, Vue, PostgreSQL database, NoSQL, Docker, AWS cloud computing, Grafana, Terraform, Jenkins, Bash, Postman, REST APIs, Cronjobs, problem-solving, verbal communication

Amazon.com, Inc

Software Development Engineer - Alexa Devices

May 2022 – January 2023

Seattle, WA

- Developed and optimized Java 8/Spring Boot microservices for Alexa Thermostat, integrating with AWS Lambda, RDS, DynamoDB, S3, and SQS, improving responsiveness by 15% and ensuring high availability during Amazon Prime Day
- Enhanced CI/CD reliability by debugging failing test cases in staging pipelines; resolved 70+ FindBugs issues and 3,000+ Checkstyle errors, improving code quality and reducing deployment failures
- Implemented "Vacation Mode" for Alexa Thermostat, enabling away scheduling and state sync across devices, improving reliability during high-traffic events
- Applied Mockito-based unit testing to strengthen reliability of backend services, and maintained test coverage for regression checks; Supported frontend QA with Node.js/React Native debugging and documentation, cutting QA cycles by 25%
- Facilitated standups and cross-trained SDEs across teams, sharing best practices in Agile/DevOps to accelerate delivery and foster team collaboration

Tech: Java 8, Spring Boot, TypeScript, MySQL, Node.js, React Native, AWS (Lambda, RDS, DynamoDB, S3, SQS, CloudWatch), C++, Redis, Git, data structures, Maven, Gradle, Mockito, Agile/Scrum, self-directed learning

Projects

Opioid Efficiency using NLP | *Python, scikit-learn, NLTK, data modeling*

- Applied **Machine Learning** and **NLP** to large-scale datasets in research, gaining familiarity with **Spark-based workflows** and distributed data processing concepts.

Pac Man World - AI Powered Game | *Python, deep learning*

- Developed a Pac Man game with autonomous Pacman and Ghost agents, leveraging bi-directional and A-star search algorithms, and achieving optimized pathfinding and gameplay
- Conducted comparative analysis on agent efficiency across A-star, DFS, and BFS algorithms using Machine Learning and visualization